

3 Energy security and the future

Enquiry question: *What might the world's energy future be?*

What students need to learn	Suggested teaching and learning
■ There is uncertainty over both global energy supply in terms of reserves (eg peak oil and gas) and demand (economic growth rates, conservation of resources, a switch to renewable sources).	■ Investigating a range of energy supply and demand, and economic growth projections to develop an awareness of uncertainty.
■ There are different responses to increasing energy demands – such as 'business as usual' reliance on fossil fuels or the adoption of alternative sources such as nuclear, or wind power. Each has costs and benefits, such as future climate change, and local opposition.	■ Weighing up the advantages and disadvantages of contrasting energy options in terms of their technical feasibility, economics and environmental impacts, including waste products.
■ Energy insecurity may lead to increased geopolitical tension and the potential for conflict, eg in the Middle East, or between gas consumers in Europe and producers in Russia, as consumers attempt to secure supplies.	■ Developing an awareness that striving for energy security within a 'business as usual' framework may increase the risk of geopolitical tension in production hotspots.
■ Meeting future energy needs in developing, emergent and developed economies while avoiding serious environmental degradation requires up-scaling of radical new approaches (conservation, recycling, reliance on renewables, carbon credits and 'green' taxation) involving difficult choices.	■ Weighing up radical policies and investigating their sustainability as well as examining the attitudes of different players towards these alternative energy futures.

3.4 Topic 2: Water conflicts

Water resources

Water, like energy, is a fundamental human need, but is not evenly distributed. Physical factors play a key role in determining the geography of surface and groundwater supplies, as does human management and mis-management of the water resource base. Increasingly demand for water, which is growing, does not match supply and this can have implications for human wellbeing. Demand for water resources comes from various users, and in addition water resources are often trans-boundary in nature.

Water conflict

The potential for conflict — both local and international — is high, and in many cases water resource use exceeds recharge capacity leading to long-term degradation. The future of water supply is in doubt in many areas, due to unsustainable use and the threat of climate change; increasingly it is already vulnerable populations who stand to suffer the most. Developing management strategies to ensure supply will require the co-operation of many different players, and changes in the way water is valued and used.

1 The geography of water supply

Enquiry question: *What is the geography of water supply and demand?*

What students need to learn	Suggested teaching and learning
■ Water supply is controlled by physical factors, such as climate, geology and surface processes; supply can be from surface or groundwater sources; fresh water supply is a finite resource.	■ Investigating the world's global fresh water supply, and developing an understanding of its link to climate zones, river systems and subsurface geology in terms of aquifers.
■ There is often a growing mismatch between water supply and demand, which can lead to water stress either locally, or across whole regions eg economic growth in the RICs such as China and India.	■ Researching trends in water use, stores and supplies and identifying areas of water stress.

■ Human activity can affect water availability, through processes such as pollution of supply, over abstraction and salt-water incursion; this can exacerbate water stress.	■ Investigating the processes that can lead to water supply deterioration in terms of both quantity and quality, and the players involved in this.
■ Access to water is often related to and controlled by wealth and poverty, especially in developing and emerging economies.	■ Exploring the link between water insecurity and poverty/wealth in countries at contrasting levels of development.

2 The risks of water insecurity

Enquiry question: *What are the potential implications of an increasingly 'water insecure' world?*

What students need to learn	Suggested teaching and learning
■ The development, extraction and use of water sources can lead to environmental and supply problems eg in the Middle Eastern or India/ Bangladesh, with severe implications for human welfare and economic activity.	■ Developing awareness that water availability is fundamental to human health and sustainable economic growth.
■ There is potential for water conflict where demand exceeds supply, and where several players use the same water system.	■ Exploring how water and its use may be a significant cause of tension and may strain relations between and within countries eg Middle Eastern or Himalayan river systems.
■ Increasingly water supply is a geopolitical issue, often with a trans-boundary dimension eg water trading along the Colorado between states of the USA, the rise in water treaties, and the political power associated with these.	■ Investigating the role of agreements and treaties between areas and surplus and deficit, and weighing up the winners and losers.
■ There are environmental and political risks of developing pathways between areas of water surplus and of deficit nationally (eg in Australia) and internationally (eg between Israel and Turkey).	■ Developing an understanding of how and why pathways between water sources and consumers are complex and why they show increasing levels of risk in regions of conflict.

3 Water conflicts and the future

Enquiry question: *What are the possible conflicts and solutions to increasing demands for water?*

What students need to learn	Suggested teaching and learning
■ Trends in water demand (improved standards of living, economic growth in industry and agriculture) and supply suggest an increasingly insecure water future for many regions, such as the Indian subcontinent. Climate change may have a significant impact in some areas.	■ Investigating projections, and the causes of their inherent uncertainties (such as the role of global warming), of future water supply and demand at global and regional scales.
■ Different players and decision makers have key roles to play in determining the future water security; their aims may conflict (water companies, environmentalists, individual, government).	■ Developing awareness that water consumers, providers and governments face difficult choices in future decades, and that further exploitation of water resources could involve political conflict.
■ There are a range of responses to current and projected demands for water such as diverting supplies and increasing storage, for example China and Brazil; or water conservation; or restoration of lost supplies, for example the Aral Sea or Long Island.	■ Investigating alternative strategies that exist for managing water supplies in future, in terms of their social, economic, political and environmental costs and benefits.
■ Technology may play a role in increasing water supply, for instance water transfer schemes and desalinisation, although this is likely to have environmental costs.	■ Exploring the role of technology, and differing access to technology, in ensuring supply and reducing water insecurity.